

DHS/MIS malaria prevalence and child mortality analysis

Summary of main results

- **Higher malaria prevalence is associated with higher post-neonatal mortality.** In the ridge-linear model, a 10 percentage-point increase in PfPR2-10 is associated with 8.8% higher post-neonatal mortality (95% CI: 5.8%–11.8%). The estimate for neonatal mortality, our negative-control outcome, is small and statistically consistent with no association. See association results and neonatal model selection.
- **After adding national vaccine coverage, a time-stable nonlinear prevalence-mortality relationship fits best.** The additive spline has lower AIC than the spline interaction (Δ AIC 1.81) and the full $\text{te}(\text{PfPR}, \text{year})$ surface (Δ AIC 11.66). The spline interaction has $p=0.078$. See model comparison.
- **Time interactions remain an important sensitivity for the historical trend, even though they are not selected.** From 2000 to 2024, estimated deaths fall by 53% with no interaction, 25% with the spline ti interaction and 32% with the full te surface. See time-interaction results.
- **Our preferred model estimates about 477,000 malaria-attributable post-neonatal deaths for the latest period, but this estimate is uncertain and is not directly comparable to IHME or WHO.** The model-parameter 95% interval is 342,000–593,000. The point estimate is about 1% higher than the IHME/GBD 2025 estimate and 10% higher than an approximate WHO under-five estimate; the interval overlaps both. The calculation evaluates the time-stable relationship using 2024 prevalence, mortality and birth inputs. See aggregate burden comparison.
- **The aggregate agreement with IHME masks large country differences.** The model is higher for DR Congo, Chad and Sudan, but lower for Uganda, Burundi, Tanzania, Ethiopia, Ghana and Rwanda. Nigeria is about 17% higher than IHME. See country-level comparison.
- **Adding WUENIC vaccine coverage changes the temporal conclusion more than the cross-sectional association.** Exact-year national Hib3, PCV-completion and rotavirus-completion coverage have no missing values in the analysis panel and enter the ridge block. The direct DHS vaccine fields remain too sparse. See covariate processing.
- **Our current bottom line is that the positive cross-sectional association remains fairly robust, while likelihood choice remains an important limitation.** Alternative samples produce similar attributable fractions, but the log-Gaussian sensitivity is positive and statistically imprecise. See sensitivity analyses and interpretation and limitations.
- **The vaccine-adjusted result brings the preferred aggregate estimate close to IHME, but does not validate either estimate.** The model measures direct and indirect malaria-associated post-neonatal mortality, whereas IHME and WHO assign deaths specifically to malaria. See interpretation and limitations.

The sections below describe the data and methods underlying these claims, followed by the detailed results and sensitivity analyses.

Analysis structure

Data selection

The unit of analysis is a DHS/MIS survey-region-year.

1. Eligible DHS and MIS Births Recodes are surveys conducted from 2000 through 2025 in sub-Saharan Africa. AIS surveys are not included.
2. Survey regions are matched to DHS first-level survey boundaries.

3. Each region receives the population-weighted MAP PfPR2-10 estimate from the survey’s calendar year. A missing 2025 MAP surface is flagged; 2024 is not silently substituted.
4. Regional all-under-five mortality and neonatal mortality are estimated from birth histories using the DHS synthetic-cohort life-table method. Post-neonatal mortality is calculated as U5MR minus NNMR. The birth-exposure denominator is retained for the count model offset.
5. The assembled validation panel contains 936 survey-region-years from 106 surveys and 35 countries. The shared primary sample requires:
 - country mean PfPR2-10 greater than 1%;
 - regional PfPR2-10 at least 1%;
 - positive finite birth exposure, U5MR, NNMR and post-neonatal mortality.
6. The resulting primary sample contains 921 region-years, 105 surveys and 34 countries, spanning 2000–2024.

Both the country-level and region-level prevalence flags remain in the analysis dataset. Sensitivity analyses separately include sub-1% regions and restrict the prevalence range to 5–40%.

Covariate processing

Twenty-seven candidate covariates are evaluated before imputation. Variables with at most 5% missingness are retained and singly imputed using the within-country median, with the overall median as a fallback. Variables with more than 5% missingness are flagged and excluded.

The 19 included covariates are:

Level	Included covariates
Survey-region	Urban residence, regional DTP3 coverage, measles vaccination, facility delivery, maternal education, exclusive breastfeeding, birth interval under 24 months, maternal age at first birth, improved water, improved sanitation and household electricity
National, exact survey year	WUENIC Hib3, PCV-completion and final-dose rotavirus coverage
National, nearest year	DTP3 coverage, log GDP per capita, health expenditure as a share of GDP, log health expenditure per capita and electricity access

The eight excluded variables are direct regional pentavalent-dose-3 coverage (55.1% missing), PCV-dose-3 coverage (57.1%), completion of the survey-specific rotavirus schedule (59.0%), household wealth (5.56%), stunting (25.2%), underweight (27.5%), wasting (27.5%) and political stability (100%).

The direct vaccine measures use standard DHS vaccination recodes: `h51–h53` for pentavalent, `h54–h56` for pneumococcal vaccine and `h57–h59` for rotavirus. They estimate weighted coverage among living children aged 12–23 months from vaccination-card or maternal-report data. These questions measure receipt, not the national introduction date, and are concentrated in newer surveys. Missing direct-DHS fields are not coded as zero.

The national series come from UNICEF’s 2025-revision WUENIC estimates: `IM_HIB3`, `IM_PCVC` and `IM_ROTAC`. They are joined by `ISO3` and exact survey year. Reported zeros are preserved; years before a country’s first estimate and countries with no series are coded zero as not-yet-introduced, with a separate status flag. Gaps after a series starts remain missing. All 936 rows have values for the three national series, so they enter the ridge block without imputation.

The local UNICEF dataflow does not contain child HIV prevalence. Its only HIV epidemiology series is a count of 10–19-year-olds living with HIV for 2010–2024, so it is not relabelled or merged as under-five prevalence. UNAIDS AIDSinfo remains the appropriate source for country-year paediatric prevalence. Historical subnational SMC coverage is not currently available as one public machine-readable panel. The source choices are documented in `R_dhs/COVARIATE_SOURCES.md`.

Proportions are logit transformed, continuous variables remain on their declared scale, and all included variables are standardised. They enter the model as one ridge-penalised matrix block; the amount of shrinkage is estimated from the data.

Main outcome model

For each survey region, the mortality rate is converted to an approximate death count using the DHS birth exposure. Models use:

- a negative-binomial likelihood with log link;
- `offset(log(exposure))`;
- a ridge penalty on the full eligible covariate block;
- country random intercepts;
- country-specific random linear PfPR slopes.

Five models are compared using maximum likelihood and AIC on the same 921 observations:

Model	Prevalence and time specification	AIC	Δ AIC
Spline without interaction	$s(\text{PfPR}) + s(\text{year})$	7112.12	0.00
Spline with ti interaction	$s(\text{PfPR}) + s(\text{year}) + \text{ti}(\text{PfPR}, \text{year})$	7113.93	1.81
Linear without interaction	$\text{PfPR} + s(\text{year})$	7120.28	8.16
Full tensor surface	$\text{te}(\text{PfPR}, \text{year})$	7123.78	11.66
Linear with interaction	$\text{PfPR} + \text{PfPR} \times \text{year} + s(\text{year})$	7128.32	16.20

The full tensor surface incorporates the PfPR main effect, year main effect and their interaction in one term and uses 11.45 effective degrees of freedom. It is more flexible than the selected additive $s(\text{PfPR}) + s(\text{year})$ decomposition but fits less well by AIC.

The selected specification is refitted with REML for post-neonatal mortality, all-under-five mortality and neonatal mortality. The linear no-interaction model is also refitted for all three outcomes to give an interpretable percentage change per 10 PfPR percentage points.

The same five-model comparison is also run independently for neonatal mortality as a diagnostic. The primary negative-control comparison nevertheless retains the post-neonatal-selected specification to avoid choosing a different functional form based on the negative-control outcome.

Main results

DHS/MIS association

From the ridge-linear summary models, each 10 percentage-point increase in PfPR2-10 is associated with:

Outcome	Change in mortality		95% CI	p-value
Post-neonatal	+8.75%	+5.77% to +11.83%		<0.001
All under-five	+6.41%	+4.12% to +8.76%		<0.001
Neonatal negative control	+0.98%	−1.16% to +3.17%		0.373

The neonatal result reduces, but does not eliminate, concern that the main association reflects broad differences in child survival between higher-prevalence and lower-prevalence settings.

Neonatal model selection

The additive spline without a time interaction also has the lowest neonatal AIC, although the preference over the linear no-interaction model is weak:

Neonatal specification	AIC	Δ AIC
Spline without interaction	6008.68	0.00
Linear without interaction	6009.64	0.97
Linear with interaction	6010.92	2.24
Full tensor surface	6012.41	3.73
Spline with τ_i interaction	6016.71	8.03

In the REML refit, the selected neonatal prevalence smooth is effectively linear (edf 1.00) and remains consistent with no association ($p=0.374$). The complete post-neonatal and neonatal comparison is in `results/dhs_rebuild/outcome_specific_aic.csv`.

Covariate associations

Ridge-standardized conditional associations

Best-fitting post-neonatal model; conditional 95% intervals for a 1 SD increase after transformation

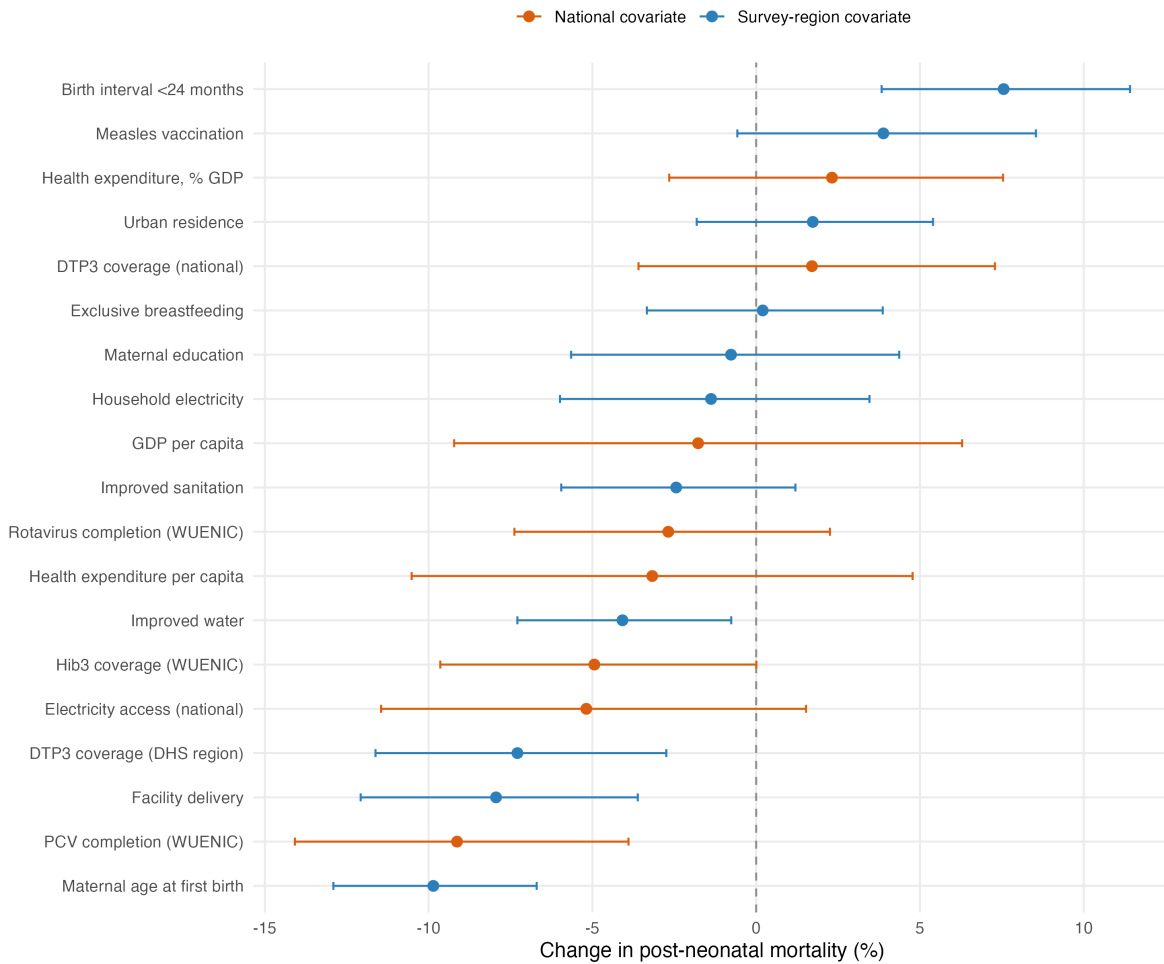


Figure 1: Ridge-standardized conditional associations for the 19 covariates in the best-fitting post-neonatal model.

The forest plot reports the percentage change in post-neonatal mortality for a one-standard-deviation increase in each transformed covariate, conditional on the complete model. The coefficients and intervals are ridge-shrunk and should not be interpreted as independent causal effects.

Attributable fraction

The selected time-stable spline estimates the following population-average fractions of post-neonatal mortality attributable to malaria relative to a 1% PfPR counterfactual:

PfPR2-10	Attributable fraction	Model-parameter 95% interval
10%	14.0%	7.7%–19.8%
30%	34.0%	25.5%–41.6%
50%	40.1%	30.5%–48.3%

Country random effects are set to zero for these population-average predictions and for the national burden estimates.

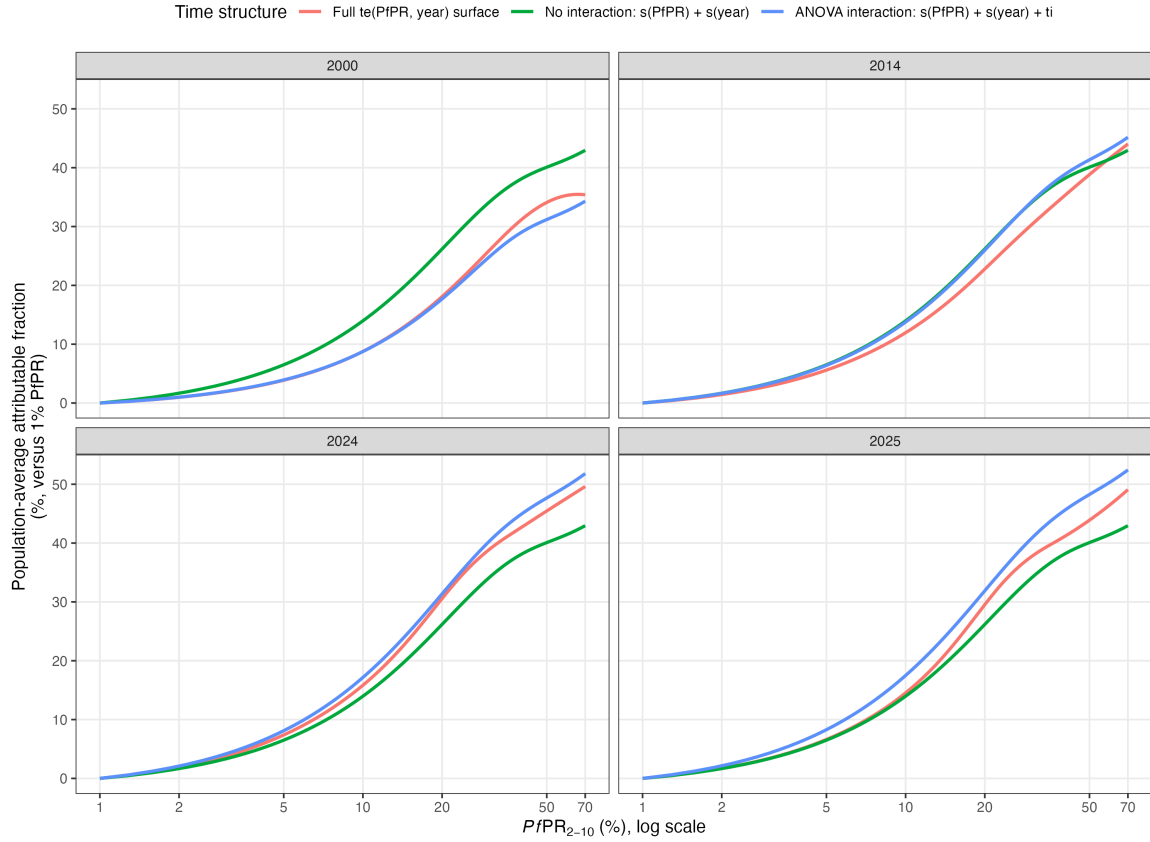


Figure 2: Attributable-fraction curves under the alternative time structures

Does the time interaction materially affect predictions?

Yes, particularly for historical trends and the latest aggregate burden, even though the additive spline is the AIC-selected specification.

At 30% PfPR, the attributable fraction is:

Model	2000	2014	2024	2025
No time interaction	34.0%	34.0%	34.0%	34.0%
Spline ti interaction	24.7%	34.1%	40.2%	40.7%
Full te surface	25.8%	30.2%	38.8%	37.7%

Applied to the same national prevalence and all-cause mortality series, the estimated post-neonatal malaria burden changes from 2000 to 2024 as follows:

Model	2000 deaths	2024 deaths	Change
No interaction	1,007,000	477,000	−52.6%
Spline ti interaction	761,000	568,000	−25.3%
Full te surface	812,000	549,000	−32.4%

This is the most important sensitivity for the manuscript’s temporal bottom line. The AIC-selected time-stable relationship implies a decline of about 53%. The alternative time-varying relationships imply smaller declines of about 25%–32%.

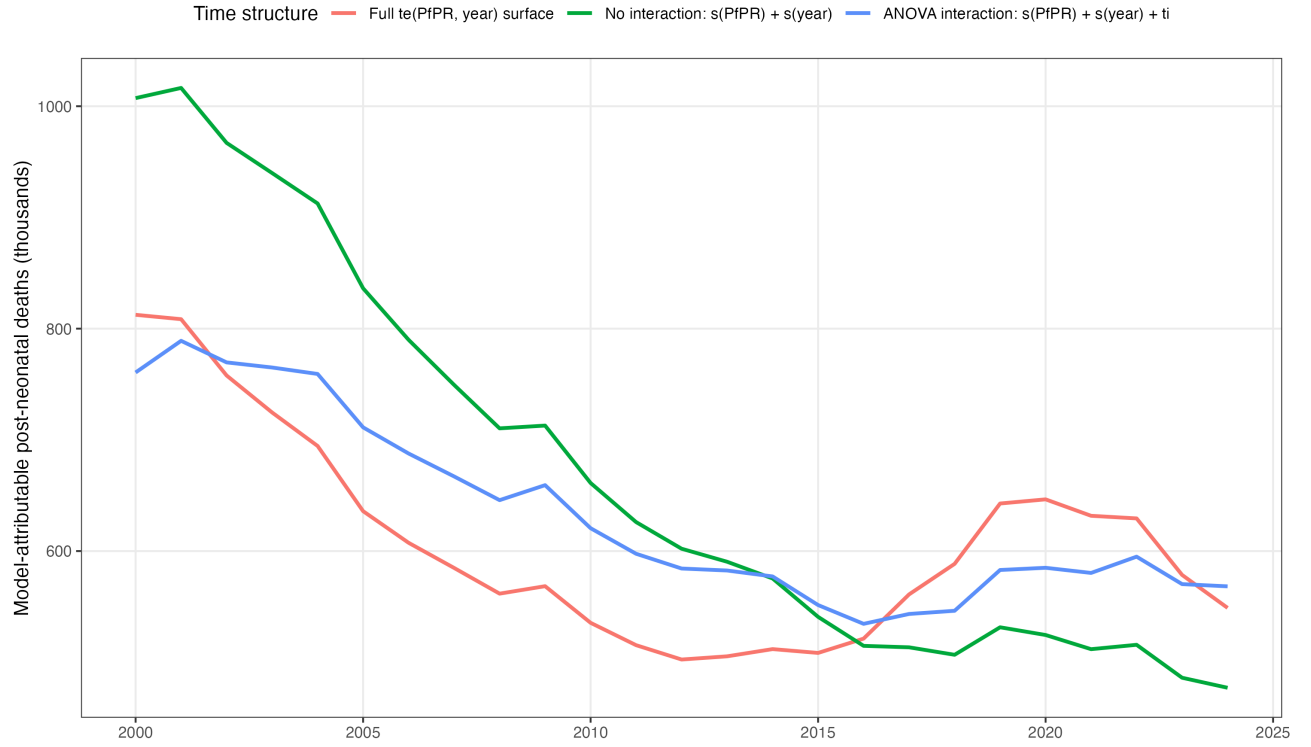


Figure 3: National burden trajectories under the alternative time structures

For the latest 2025-surface comparison, the estimates are:

Model	Estimated deaths	Model-parameter 95% interval
No interaction	477,000	342,000–593,000
Spline ti interaction	577,000	409,000–723,000
Full te surface	530,000	188,000–796,000

Thus the spline interaction raises the latest point estimate by about 100,000 deaths, or 21%, compared with the selected no-interaction model. The full **te** estimate lies between the two.

Other sensitivity analyses

Alternative samples

All sample sensitivities retain a positive nonlinear prevalence relationship:

- Primary sample: AF at 30% PfPR = 34.0%.
- Include regions below 1% PfPR: 33.7%.
- Restrict PfPR to 5–40%: 34.4%.
- Complete cases only: 34.7%.

These choices do not materially change the cross-sectional attributable fraction.

Model structure and covariate blocks

- Removing the country random PfPR slope worsens fit by about 13.0 AIC units.
- Regional covariates alone fit about 17.5 AIC units worse than the full regional-plus-national block.
- National covariates alone fit very poorly (about 225 AIC units worse).
- Changing the PfPR spline basis to **k=4** improves fit by 0.9 AIC units; **k=8** fits about 18.2 units worse. AF at 30% ranges from 32.4% to 34.4% across these basis choices.
- Across structural sensitivities, AF at 30% ranges from 32.4% to 44.8%; the upper value occurs in the national-covariate-only model, which fits poorly.

Alternative likelihood

The log-Gaussian rate sensitivity is less precise than the primary negative-binomial count model:

- ridge-linear post-neonatal estimate: +5.75% per 10 PfPR points (95% CI -1.73% to $+13.80\%$; **p=0.136**);
- selected additive-spline prevalence term: **p=0.136**;
- the no-interaction spline remains the selected likelihood sensitivity.

This weakens certainty about the exact effect size and the time interaction, but the point estimate remains positive.

Latest national burden comparison

Aggregate bottom line

The closest available comparison is not perfectly contemporaneous:

- **Model:** surface evaluated at 2025 using 2024 MAP PfPR, IGME mortality and live births for 43 matched sub-Saharan African countries.
- **IHME:** 2025 under-five malaria deaths.
- **WHO:** WMR 2025 estimates for 2024. WHO reports all-age country deaths, so the under-five comparator is an approximate 75% share.

Source	Deaths
Selected time-stable prevalence/all-cause model	477,000
IHME/GBD SSA under-five, 2025	470,000
WHO African-region all-age, 2024	579,000
WHO African-region under-five proxy, 2024	434,000

WHO’s 2025 annex explicitly supplies country estimates through 2024, not 2025. WHO also states that children under five account for approximately 75% of malaria deaths in the African Region. Sources: WMR 2025 Annex 4H, WHO malaria fact sheet, and the WHO GHO **MALARIA_EST_DEATHS** country indicator.

Countries with the largest differences from IHME

The table below uses the AIC-selected time-stable spline. “Very different” is defined in the generated results as at least a two-fold difference and an absolute difference of at least 1,000 deaths.

Country	Model	IHME 2025	Model/IHME	Interpretation
DR Congo	108,800	68,300	1.59	Largest absolute excess: +40,500
Nigeria	176,100	150,500	1.17	Model higher by about 25,600
Chad	12,100	5,500	2.21	Model substantially higher
Sudan	3,900	1,400	2.85	Model substantially higher
Uganda	13,200	28,500	0.46	Model lower by about 15,300
Burundi	3,800	15,700	0.24	Model substantially lower
Tanzania	5,000	10,700	0.47	Model substantially lower
Ethiopia	3,900	9,300	0.42	Model substantially lower
Ghana	3,500	7,300	0.48	Model substantially lower
Rwanda	300	2,500	0.11	Model substantially lower

Against the approximate WHO country-level under-five proxy, the clearest two-fold discrepancies are:

- higher under the model: DR Congo;
- lower under the model: Tanzania, Ethiopia, Ghana, Sudan, Rwanda, Kenya and Madagascar.

DR Congo and Nigeria account for most of the model’s upward difference from IHME, but this is largely offset by lower estimates for the countries listed above. This is why the aggregate model and IHME totals are close despite large country-level differences.

Interpretation and limitations

The comparison is not a validation against a common estimand:

- the prevalence/all-cause model is intended to include direct and indirect malaria-associated post-neonatal deaths;
- IHME and WHO estimate deaths assigned specifically to malaria;
- the model uses population-average effects and ignores country random effects for national extrapolation;
- 2025 exposure and all-cause mortality inputs are unavailable, and WHO’s latest estimates are for 2024;
- the WHO country under-five comparison applies a regional 75% age-share assumption to every country.

Our current interpretation is that the data provide fairly robust evidence of a positive cross-sectional association between malaria prevalence and post-neonatal mortality after adjustment for national vaccine rollout. The preferred aggregate estimate is now close to IHME, but it is not a validation against a common estimand. The selected time-stable model implies a decline of about 53% since 2000; time-varying sensitivities imply smaller declines of about 25%–32%.